

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-07 15:45:41

Training Start: 2025-05-07 15:33:39

Training End: 2025-05-07 15:43:32

Training Duration: 9.88 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: relu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 3.3492

MSE: 15.0015

RMSE: 3.8732

R2 Score: -1.6226

Multi-Step Forecast:

Forecast Steps: 20

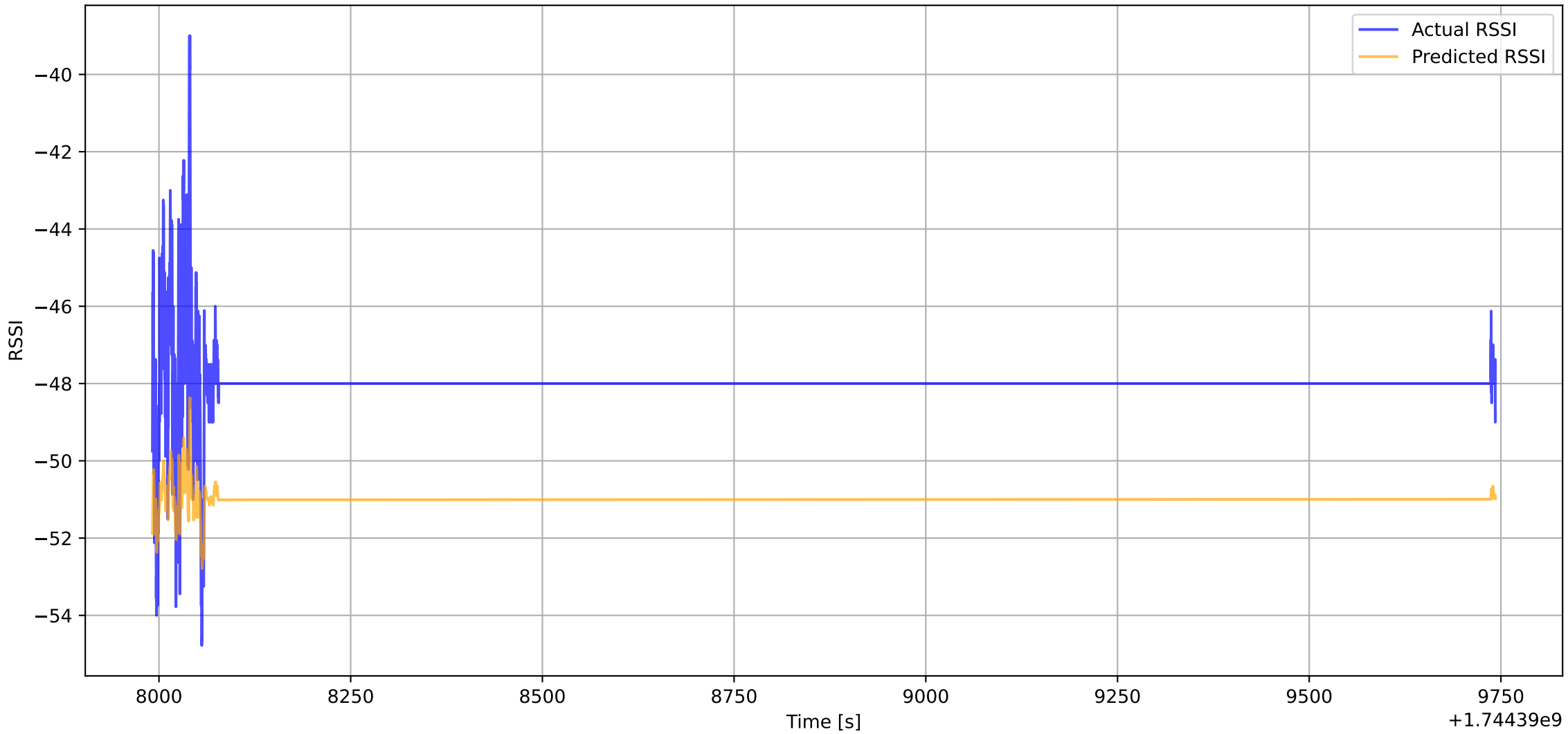
MAE: 0.1217

MSE: 0.0188

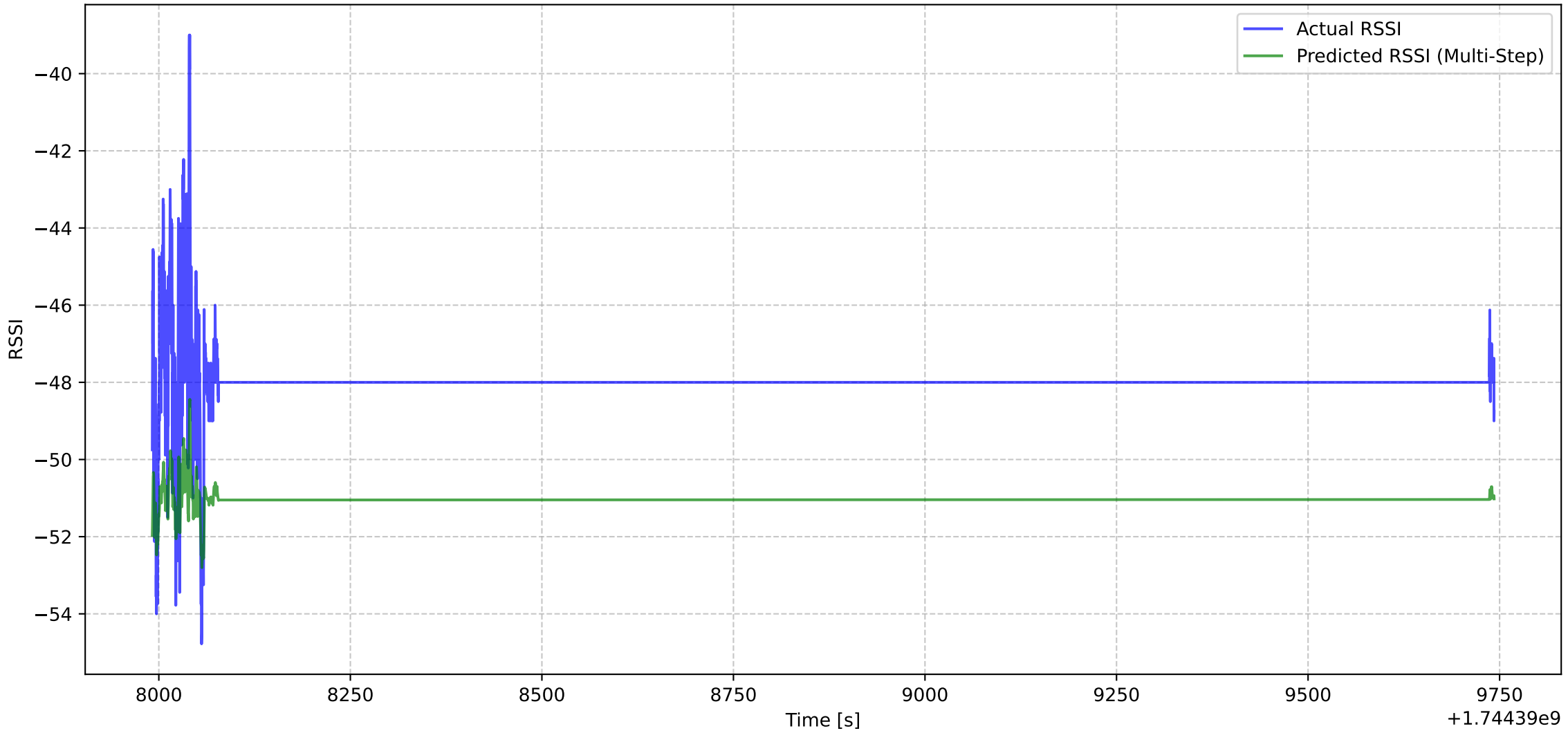
RMSE: 0.1370

R2 Score: -1.3310

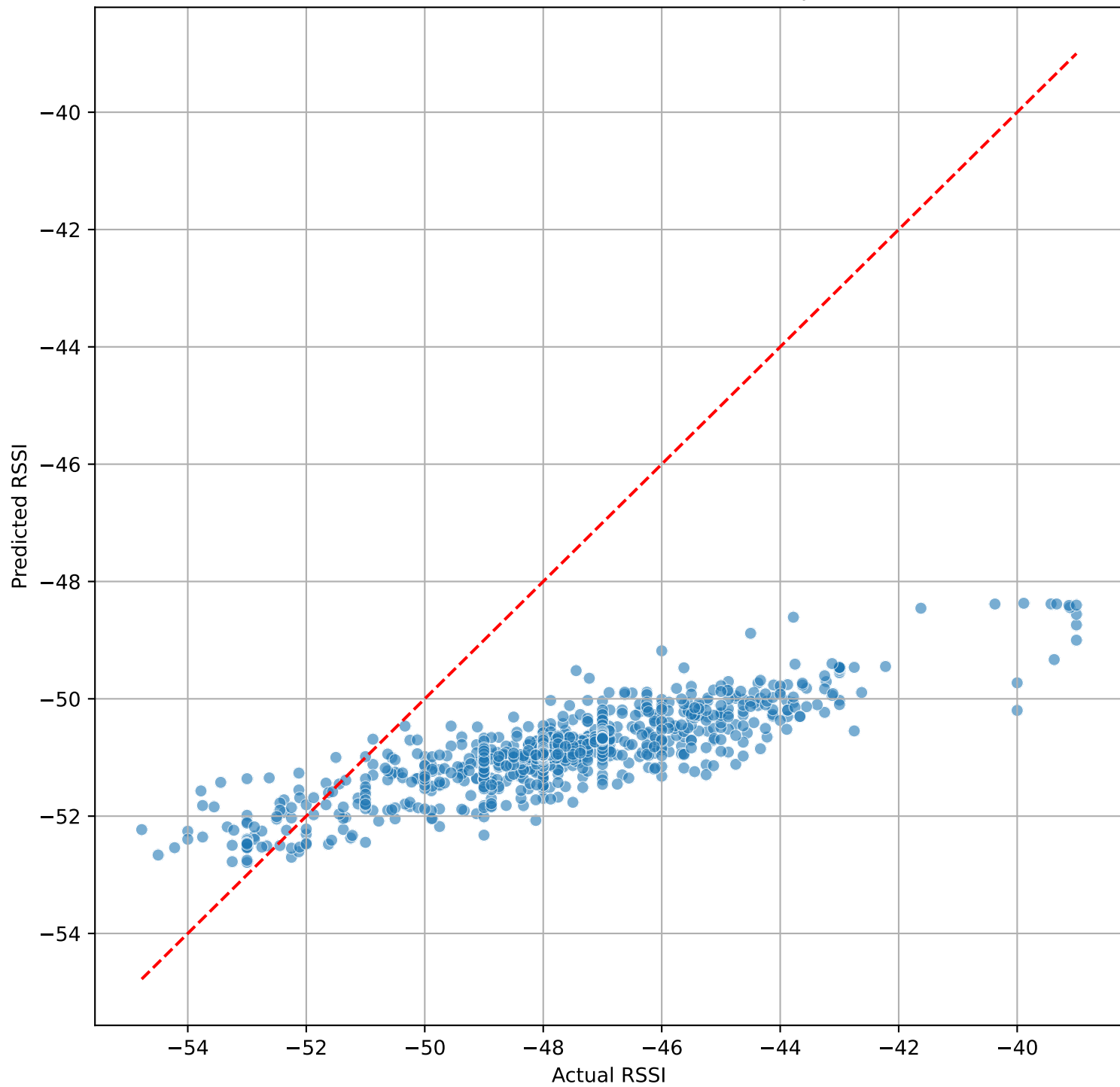
Actual vs Predicted RSSI (First Step)



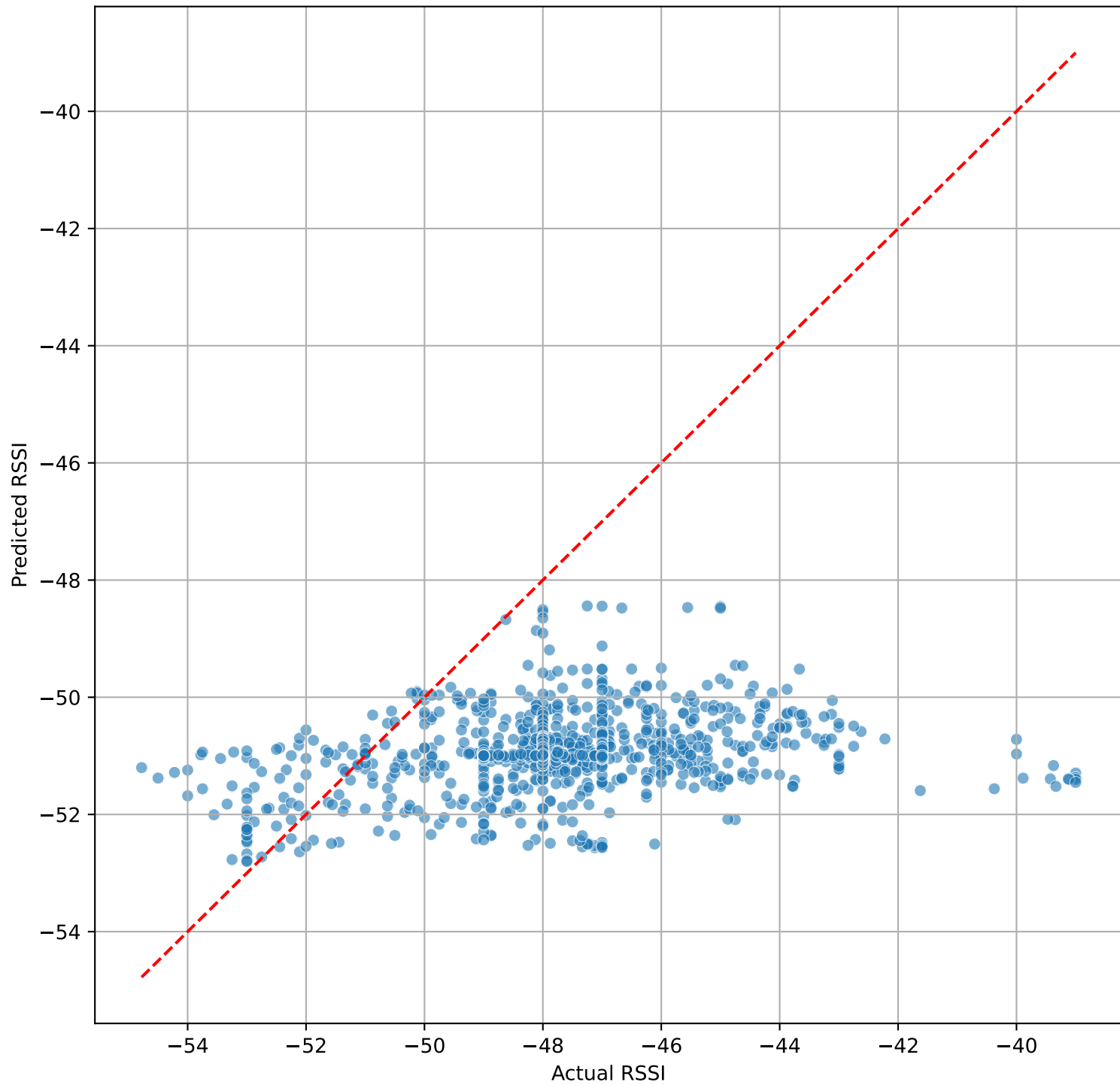
Actual vs Predicted RSSI (Multi-Step (step 20))



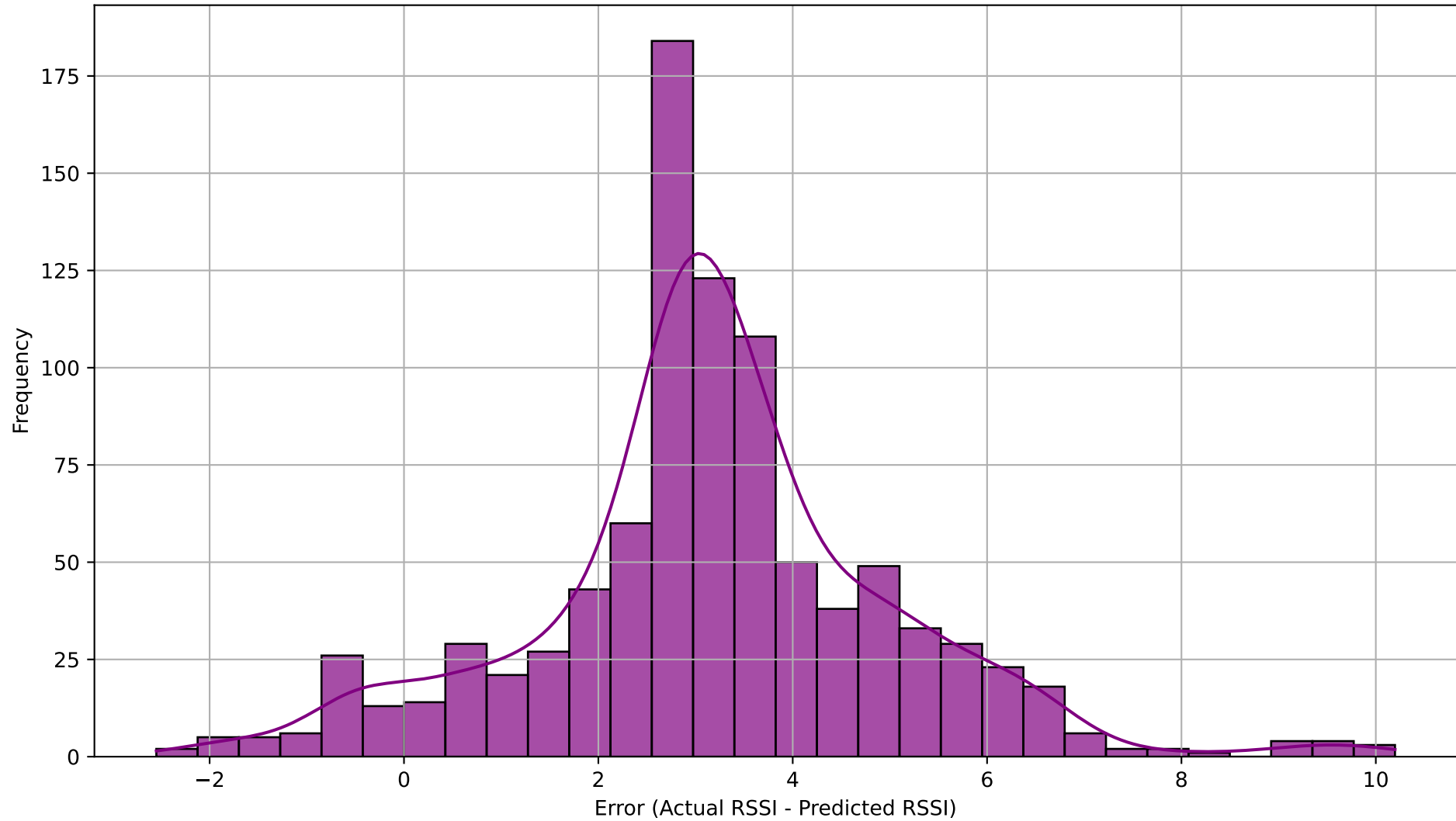
Actual vs Predicted RSSI (First Step)



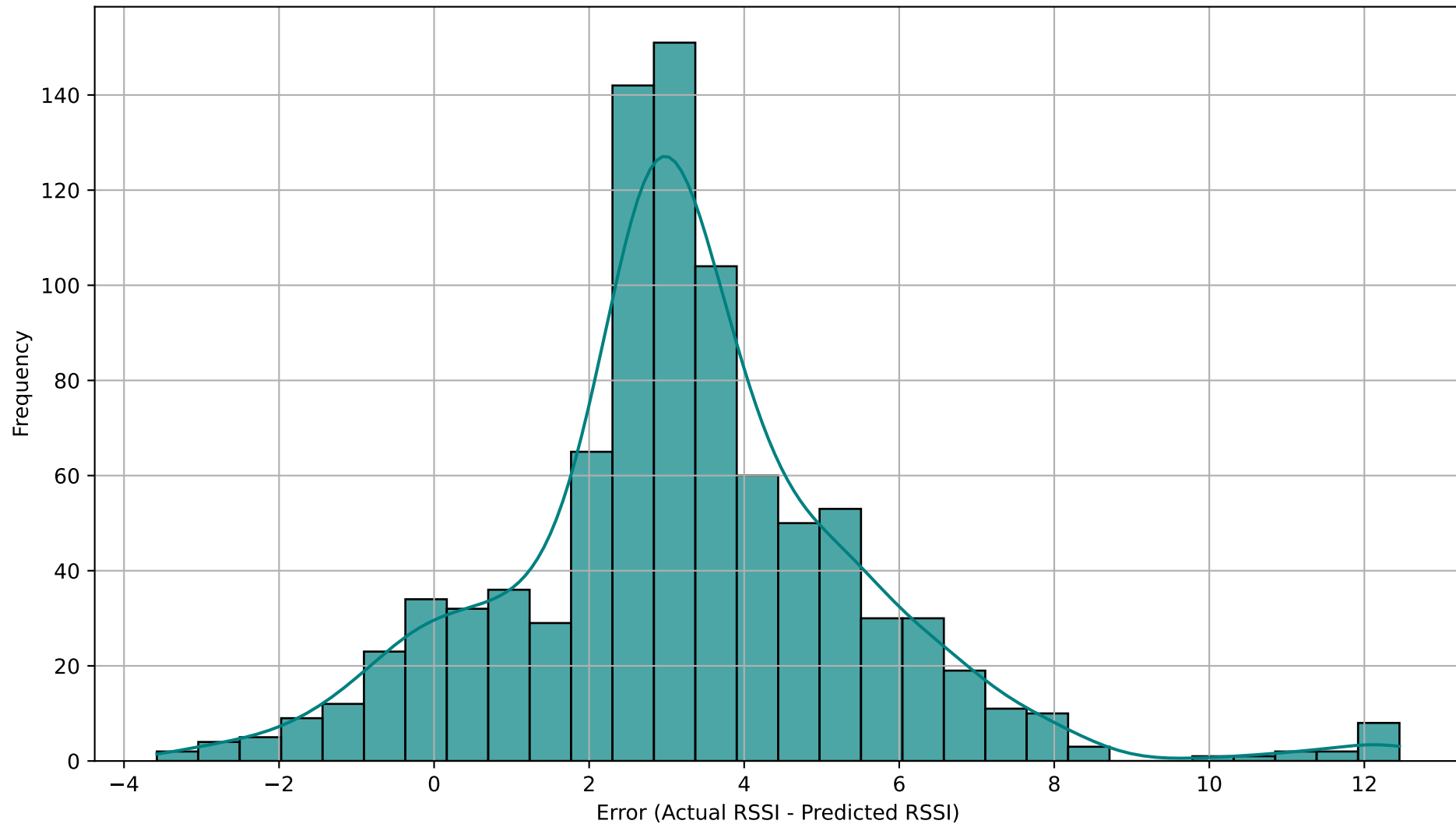
Actual vs Predicted RSSI (Last Step - step 20)



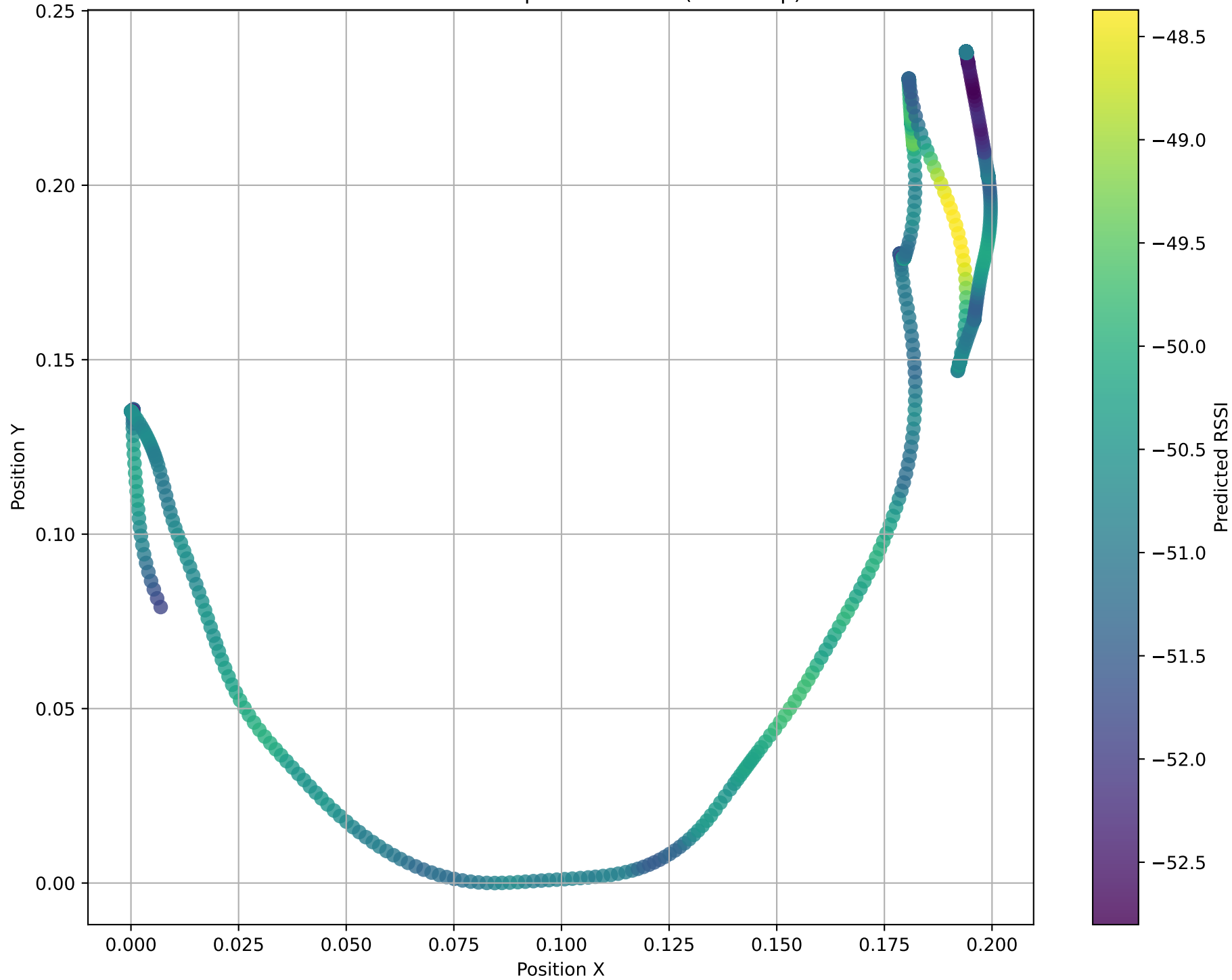
Distribution of Prediction Errors (First Step)



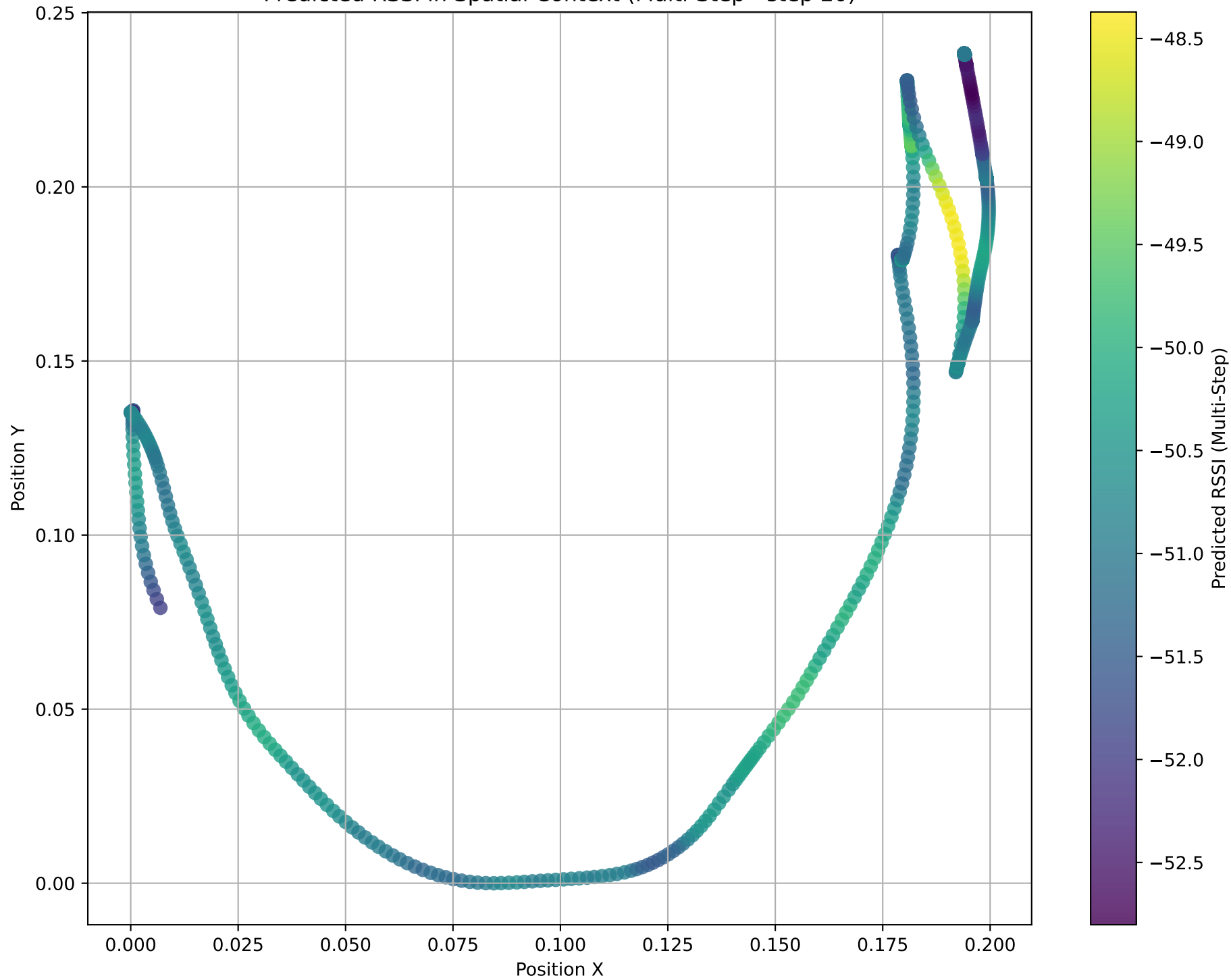
Distribution of Prediction Errors (Last Step - step 20)



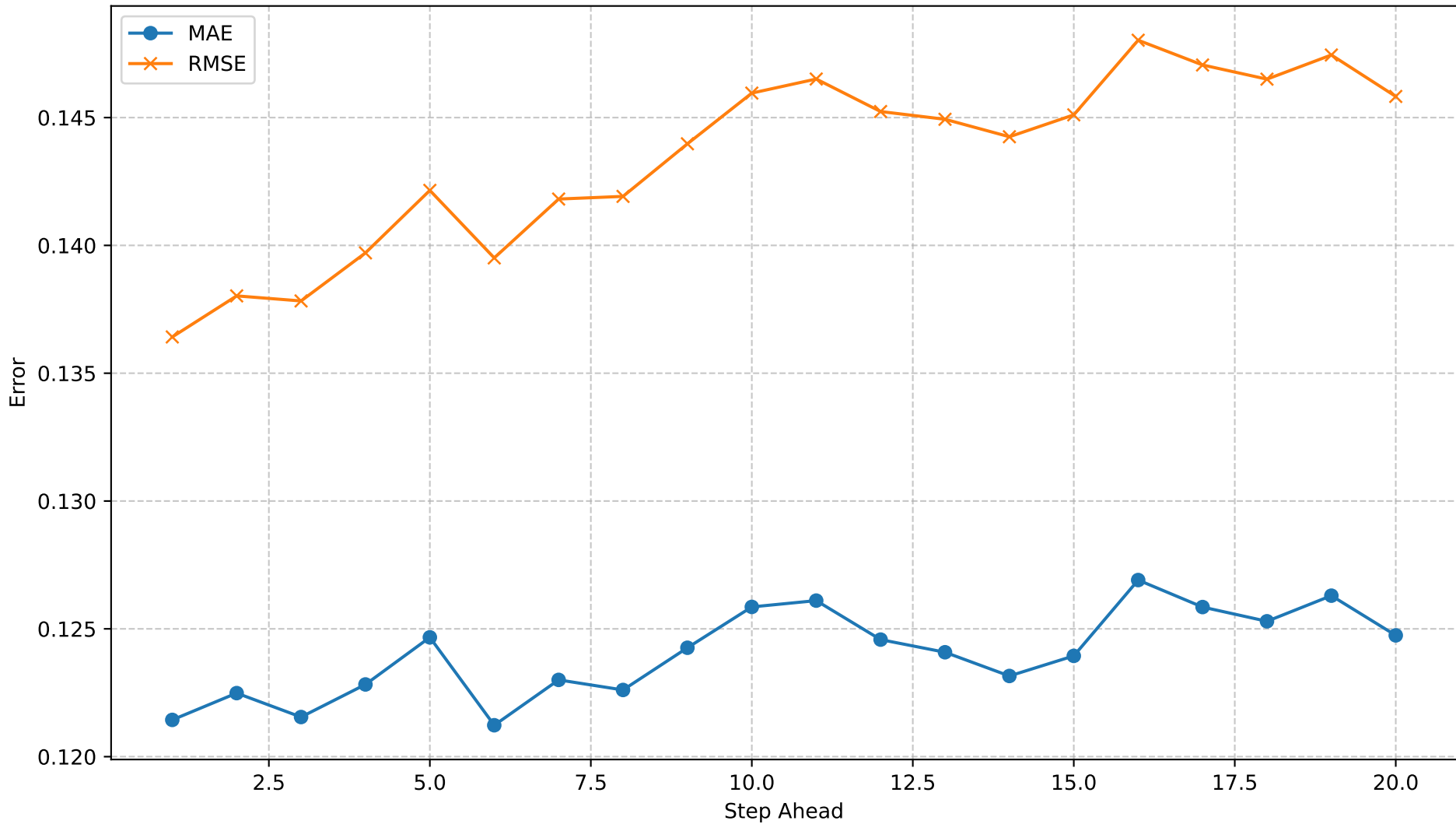
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.12 / 0.14
- Mediana : 0.12 / 0.14
- 20º passo : 0.12 / 0.15

Crescimento do erro: 2.7% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 16 (0.13), 19 (0.13), 11 (0.13)

Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.